

TP-CAN-2I — Programmable Dual-CAN Inline Interface

Engineering Specification

Revision A · 2026-06-01

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Tester Present — Programmable CAN Interface

Master Engineering Specification

Product	Programmable Dual-CAN Inline Interface (CAN / Serial / USB / Wireless)
Model	TP-CAN-2I (Tester Present Specialist Automotive Solutions)
Document	Master Engineering Specification
Revision	A (2026-05-31)
Author	Jack Leighton — Tester Present Specialist Automotive Solutions
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Origin	Designed in Australia

1. Product Overview

The TP-CAN-2I is a ruggedised, automotive-grade **inline CAN-bus interface** that installs between a vehicle harness (upstream) and an ECU/module (downstream). It can **monitor, pass-through, filter, and optionally rewrite** CAN traffic flowing between the two network segments — a configurable "man-in-the-middle" for diagnostics, emulation, data-logging, gateway/translation, and bench development.

It is built around an **ESP32-S3** providing Wi-Fi (telemetry / OTA) and Bluetooth LE (configuration), with **two fully galvanically-isolated CAN FD channels**, a 3.3 V UART service port, and native USB-C for programming via the Arduino IDE or the Tester Present Windows configurator.

1.1 Primary Use Cases

- Inline CAN man-in-the-middle (intercept / modify / inject between vehicle and module)
- Dual-bus gateway / protocol translation (CAN-A ↔ CAN-B, different bit-rates)
- High-rate CAN/CAN-FD data logger with wireless off-load
- ECU emulation / residual-bus simulation on a bench
- Field reflash / configuration tool (talks to a target over CAN via J2534)

1.2 Design Pillars (from product brief & concept renders)

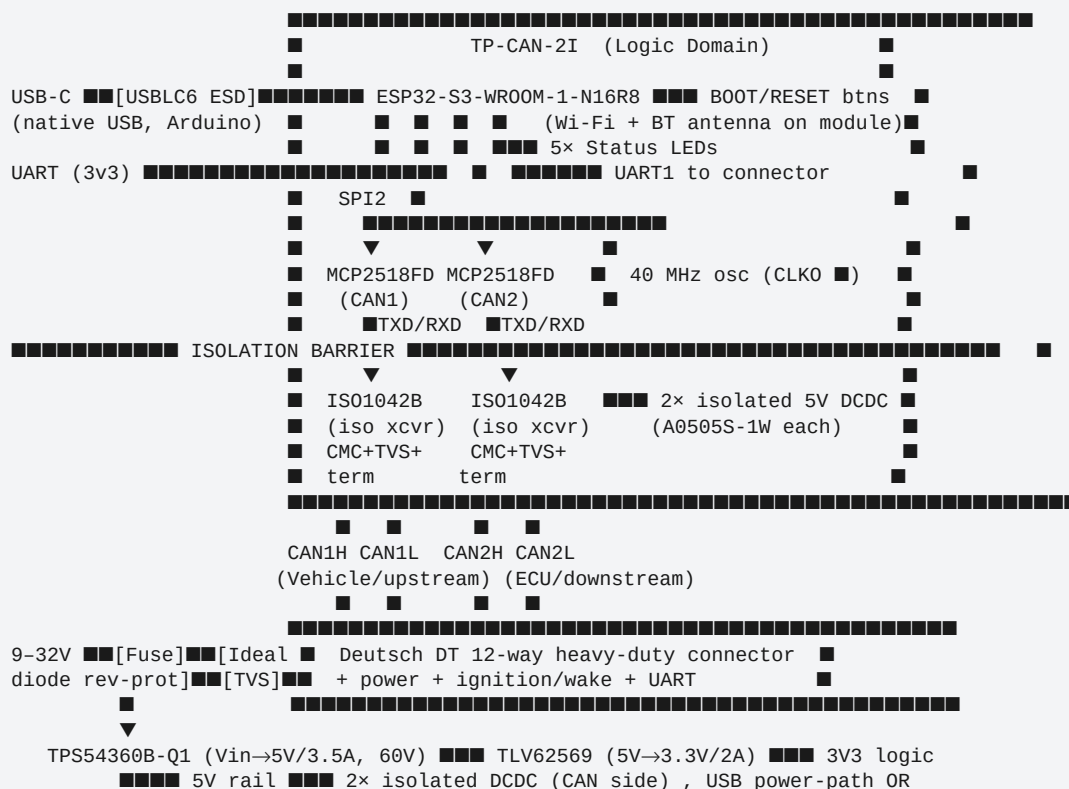
1. **Rugged & reliable** — automotive-grade components, sealed enclosure, heavy-duty connector.
2. **Wireless** — Wi-Fi for updates & telemetry, Bluetooth LE for configuration.
3. **Secure & isolated** — galvanic isolation between the two CAN sides for safety.
4. **Easy access** — USB-C programming and diagnostics; Arduino-IDE compatible.

2. Key Specifications

Parameter	Specification
MCU	Espressif ESP32-S3-WROOM-1-N16R8 (dual-core LX7 @ 240 MHz, 16 MB flash, 8 MB PSRAM)

Parameter	Specification
Wireless	Wi-Fi 802.11 b/g/n 2.4 GHz + Bluetooth 5 LE (on-module PCB antenna)
CAN channels	2 × independent, galvanically isolated CAN FD (ISO 11898-2)
CAN controller	2 × Microchip MCP2518FD (external SPI controllers, 40 MHz)
CAN transceiver	2 × TI ISO1042B isolated FD transceiver (± 70 V bus fault, 5 Mbit/s)
CAN bit-rates	Classic CAN 2.0B 125 k–1 Mbit/s; CAN FD arbitration ≤ 1 Mbit/s, data ≤ 5 Mbit/s
CAN isolation	≥ 3 kV _{RMS} channel-to-logic and channel-to-channel (reinforced option 5 kV _{RMS})
Serial	1 × 3.3 V UART (TX/RX) on connector + internal service header
USB	USB-C 2.0 Full-Speed, native (ESP32-S3 USB-OTG), ESD protected
Programming	(a) USB-C + Arduino IDE; (b) Tester Present app over USB-serial; (c) over CAN via J2534; (d) over UART serial
Input voltage	9–32 V DC nominal (supports both 12 V and 24 V systems); survives load-dump transients (clamped 53 V)
Input protection	Fuse + reverse-polarity (ideal-diode) + TVS (ISO 7637-2 pulse tolerant) + π -filter
Quiescent / sleep	< 200 μ A in deep-sleep (ignition-wake), typical run ≈ 0.6 – 1.4 W
Indicators	5 × LED: PWR, Wi-Fi, CAN1, CAN2, STATUS
Connector	Deutsch DT series, 12-way , sealed (IP67-mated), 13 A/contact
Enclosure	Sealed black SLA/ABS-like, IP65 target, 4 × heavy-duty mounting ears (brass inserts)
Operating temp	-40 °C ... $+85$ °C (component-rated); enclosure -20 °C ... $+70$ °C
PCB	4-layer FR-4, 1.6 mm, ENIG, isolation slots between CAN domains
Dimensions (PCB)	$\approx 70 \times 50$ mm (target)
Dimensions (case)	$\approx 95 \times 70 \times 32$ mm excl. connector & mounting ears

3. System Block Diagram



4. Electrical Architecture

4.1 Power Tree (12 V / 24 V capable)

Stage	Part	Function
Input fuse	3 A slow-blow (1206 / Nano2)	over-current / fault isolation
Reverse-polarity	LM74700-Q1 ideal-diode ctrl + N-FET (CSD18532)	low-loss reverse protection
Transient clamp	SMCJ33A TVS (uni) + π -filter (ferrite + bulk caps)	ISO 7637-2 load-dump / surge
Pre-regulator	TPS54360B-Q1 (4.5–60 V in, 3.5 A) → 5.0 V	wide-Vin buck, survives 24 V load-dump
Logic regulator	TLV62569 (5 V → 3.3 V, 2 A)	ESP32-S3 + controllers + logic
CAN1 bus supply	A0505S-1W isolated DCDC (5 V→5 V iso)	ISO1042B VCC2, isolated domain 1
CAN2 bus supply	A0505S-1W isolated DCDC (5 V→5 V iso)	ISO1042B VCC2, isolated domain 2
USB power-path	LM66100 ideal diode (USB 5 V → 5 V rail)	bench-power from USB without back-feed

Rail budget (typical): 3.3 V @ ≤500 mA (ESP32 Wi-Fi bursts) + controllers/LEDs ≈ 150 mA; 5 V rail feeds 2× 1 W isolated converters (≈ 200 mA each in) + USB. Pre-reg sized 3.5 A for margin.

4.2 Microcontroller

- **ESP32-S3-WROOM-1-N16R8**: native USB-OTG (no USB-UART bridge required), dual-core, 16 MB flash / 8 MB PSRAM for large MITM buffers and OTA dual-bank.
- Octal-PSRAM variant occupies GPIO33–37 internally — **those GPIO are reserved, do not use.**
- Antenna keep-out per module datasheet must overhang the PCB edge with no copper beneath.

4.3 Dual CAN FD (Isolated)

- Two **MCP2518FD** SPI CAN-FD controllers share **SPI2 (FSPI)** with independent CS + INT.
- One **40 MHz** oscillator clocks controller 1; its **CLKO** output daisy-chains controller 2.
- Each controller's TXD/RXD crosses the isolation barrier through an **ISO1042B** transceiver.
- Each transceiver bus side is powered by its **own isolated 5 V** supply → channel-to-channel **and** channel-to-logic galvanic isolation (true MITM safety).
- Bus protection per channel: **common-mode choke** (TDK ACT45B) + **CAN TVS** (PESD2CAN) + **split termination** ($2 \times 60 \Omega + 4.7 \text{ nF}$ to isolated bus GND) + jumper-selectable 120Ω .

4.4 USB / Programming

- USB-C receptacle (16-pin, with mounting tabs), **CC1/CC2 5.1 kΩ** pull-downs (UFP).
- **USBLC6-2SC6** ESD array on D+/D–.
- Native USB connects to ESP32-S3 GPIO19 (D–) / GPIO20 (D+).
- **BOOT (GPIO0) + RESET (EN)** tactiles for manual flashing; auto-reset supported by native USB.

4.5 Serial UART

- **UART1** (GPIO17 TX / GPIO18 RX), 3.3 V logic, brought to both the connector and an internal 4-pin service header (TX, RX, GND, 3V3).

4.6 Status Indicators

LED	Colour	Driven by	Meaning
PWR	Green	Hard-wired to 3V3	Power good
Wi-Fi	Blue	GPIO38	Wi-Fi/BT activity
CAN1	Amber	GPIO39	Vehicle-side bus traffic
CAN2	Amber	GPIO40	ECU-side bus traffic
STATUS	Blue	GPIO41	System/MITM state (heartbeat/error codes)

5. ESP32-S3 Pin Allocation

Function	GPIO	Notes
USB D–	19	native USB (fixed)
USB D+	20	native USB (fixed)
SPI2 SCK	12	to both MCP2518FD

Function	GPIO	Notes
SPI2 MOSI (SDI)	11	to both MCP2518FD
SPI2 MISO (SDO)	13	from both MCP2518FD
CAN1 CS	10	MCP2518FD #1 chip-select
CAN2 CS	14	MCP2518FD #2 chip-select
CAN1 INT	4	MCP2518FD #1 interrupt (input)
CAN2 INT	5	MCP2518FD #2 interrupt (input)
UART1 TX	17	3.3 V serial to connector/header
UART1 RX	18	3.3 V serial to connector/header
LED Wi-Fi	38	
LED CAN1	39	
LED CAN2	40	
LED STATUS	41	
BOOT	0	strapping — tactile to GND
Ignition/Wake sense	16	via divider+TVS; RTC-capable for deep-sleep wake
Reserved	26–37	SPI flash + octal PSRAM (N16R8) — DO NOT USE
Avoid as outputs	0, 3, 45, 46	boot strapping pins

6. Heavy-Duty Connector — Deutsch DT 12-Way

External harness connector: **Deutsch DT** series, 12-way, sealed (IP67 mated), 13 A/contact, 14–16 AWG capable. Enclosure-wall flange receptacle → short internal loom → polarised locking board header (Molex Micro-Fit 3.0, 12-ckt) on the PCB ("ECU-style" serviceable interface).

Pin	Signal	Domain	Wire (AWG / colour)
1	+VBAT (9–32 V power in)	Power	16 / Red
2	GND (power / chassis)	Power	16 / Black
3	IGN / WAKE (switched +)	Logic	18 / White
4	CAN1-H (Vehicle / upstream)	Iso domain 1	18 / Yellow (twisted pair)
5	CAN1-L (Vehicle / upstream)	Iso domain 1	18 / Green (twisted pair)
6	CAN1-GND (vehicle bus ref/shield)	Iso domain 1	18 / Bare/Drain
7	CAN2-H (ECU / downstream)	Iso domain 2	18 / Orange (twisted pair)
8	CAN2-L (ECU / downstream)	Iso domain 2	18 / Blue (twisted pair)
9	CAN2-GND (ECU bus ref/shield)	Iso domain 2	18 / Bare/Drain
10	UART-TX (3.3 V out)	Logic	22 / Grey
11	UART-RX (3.3 V in)	Logic	22 / Violet
12	UART-GND (logic ref)	Logic	22 / Brown

***Note:** CAN-GND pins (6, 9) reference each isolated transceiver's bus-side ground. For a true MITM in a single-chassis vehicle they may be left unterminated or tied to local chassis at each node; on a bench they provide the isolated return.*

Connector part numbers and the harness build procedure are in [/Harness/Connector_and_Harness.md](#).

7. Isolation Strategy

- **Logic domain:** ESP32-S3, both MCP2518FD controllers, USB, power regulators, LEDs.
- **Isolated domain 1 (CAN1):** ISO1042B bus side + its isolated 5 V supply.
- **Isolated domain 2 (CAN2):** ISO1042B bus side + its isolated 5 V supply.
- Barrier carried by the ISO1042B reinforced isolation and the isolated DCDC transformers.
- **PCB:** isolation **slots milled** under each ISO1042B and DCDC, ≥ 4 mm creepage logic-to-bus and bus-to-bus; no copper, no silkscreen, no traces crossing the barrier except through the rated parts. Maintain ≥ 8 mm spacing between the two CAN domains where practical.

8. Mechanical Envelope

- Two-part sealed enclosure (base + lid) with perimeter **gasket groove** (O-ring cord or cured-in-place gasket), IP65 target.
- **4 heavy-duty mounting ears** with **brass heat-set inserts** / steel eyelets for M5 bolts.
- Deutsch DT flange receptacle on one end face; **sealed USB-C service port** (rubber bung) on an adjacent face.
- PCB on **4 standoffs/bosses** with M3 brass inserts; 1.5 mm clearance to lid for components.
- **LED light-pipes** (5) through the lid, or a clear window strip.
- Full model + drawing: [/Enclosure/](#).

9. Compliance & Environment (targets)

Item	Target
Conducted transients	ISO 7637-2 pulses 1, 2a, 2b, 3a, 3b (clamped front-end)
Supply ranges	ISO 16750-2 (12 V & 24 V systems), load-dump survival
ESD	IEC 61000-4-2 ± 8 kV contact on exposed I/O & USB
CAN PHY	ISO 11898-2:2016 (CAN FD)
Ingress	IP65 (enclosure), IP67 (mated connector)
RoHS / REACH	Compliant component selection

10. Programming & Configuration Interfaces

Path	Transport	Use	Tool
Firmware flash	USB-C native	Load/replace firmware	Arduino IDE (ESP32-S3 board)

Path	Transport	Use	Tool
Firmware flash	USB-C native	Load/replace firmware	Tester Present app (esptool backend)
Configuration	USB-serial (CDC)	Set CAN bit-rates, MITM rules, filters, Wi-Fi creds	Tester Present app
Configuration	CAN via J2534	Program/config over the bus	Tester Present app (J2534 PassThru)
Configuration	UART serial	Headless config	Tester Present app / terminal
Live / OTA	Wi-Fi	Telemetry, OTA update	Tester Present app / web
Config	Bluetooth LE	Mobile/field config	(future)

The on-the-wire configuration protocol is defined in the firmware and mirrored by the Windows app — see [/Firmware](#) and [/Software](#).

11. Deliverables Map

Folder	Contents
/Docs	This spec, datasheet, design notes
/Hardware/KiCad	KiCad project (schematic, PCB, netlist)
/Hardware	BOM.csv , fabrication notes
/Enclosure	OpenSCAD case (base + lid), STL output, drawing
/Harness	Deutsch DT pinout + harness build document
/Firmware	ESP32-S3 Arduino firmware
/Software	.NET 4.8.1 WinForms configurator (J2534 + serial)

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